

NAME OF STUDENT:

STUDENT NUMBER:

SUBJECT:

APPLICATIONS DEVELOPMENT THEORY IV
(APT470S)

Full time and Part time

START DATE:

22 JUNE 2021

OPEN UNTIL:

22 JUNE 2021

DURATION:

2 HOURS

105 min Writing
15 min Upload

**TOTAL
MARKS:**

60



**Cape Peninsula
University of Technology**
creating futures

**DEPARTMENT OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATICS AND DESIGN**

COURSE(S):

ADVANCED DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY:
APPLICATIONS DEVELOPMENT (ADICTA)

EXAMINER:

MS G KHAN

MODERATOR:

MR T MAKHURANE

EXTERNAL MODERATOR:

DR M NORMAN

GENERAL INSTRUCTIONS

- This is an individual test which should solely be done by you.
- The test is a **timed assessment** of 105 minutes in length. One hour and 45 minutes have been allocated for writing the test and 15 minutes have been allocated as reading and upload time. Students are required to monitor the time as the test will automatically terminate once the time has expired.
- This test **counts 30%** of your final mark for this subject and covers **work from semester 2** as per the updated topic list and/or Subject Guide.
- This assessment requires a working device, a reliable internet connection and data/bandwidth for the duration of the assessment. Therefore this test will be **available on Blackboard** for 2 hours only.
- Students are required to **complete the test in one sitting**.
- Students who experience technical difficulties during their test session are required to email me a screen shot of the issue immediately to khang@cput.ac.za so that measures can be taken to address that problem.
- Students are reminded to do their own work. CPUT's assessment rules and regulations will apply to any form of plagiarism or academic dishonesty, as per the student handbook.

Scenario for the ***Online Sentiment Based Movie Rating System***

As the software manager of a consultancy you are responsible for the design and development of an online movie rating website which allows users to rate and comment on movies. These ratings are provided as input to the website. The System Admin then checks reviews, critic's ratings and displays an online rating for each movie. Users should be able to post and store movie reviews which the system then analyzes for user sentiments located in each comment. In this initial version these checks will only be conducted on English content in the sentiment library. The system will break up the user comments to check for sentiment keywords before predicting the user sentiment associated for the comment. Once keywords are found it associates the comment with a sentiment score. All comments are gathered for each movie and an average rating score is calculated. Such a score is generated for every movie in the system. The system also sorts the scores and displays a top ten list of movies from the analysis which should be calculated automatically daily.

QUESTION 1 (10 MARKS)

Consider the excerpt of the "Manifesto for Agile Software Development" below and discuss situations where the four "values" mentioned could present challenges for the software team. "We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value: <i>Individuals and interactions</i> over processes and tools <i>Working software</i> over comprehensive documentation <i>Customer collaboration</i> over contract negotiation <i>Responding to change</i> over following a plan That is, while there is value in the items on the right, we value the items on the left more."	10
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QUESTION 2 (10 MARKS)

The human aspects in software engineering has become increasingly important. Explain why this is a concern and potential ways in which this issue could be addressed.	10
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QUESTION 3 (10 MARKS)

"Understanding the requirements of a problem is among the most difficult tasks that face a software engineer." Explain a) why a software engineer may find it difficult to develop a clear understanding of user requirements? b) what can the software engineer do to offset this issue?	5 5
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QUESTION 4 (10 MARKS)

Carefully read the scenario for an Online Sentiment Based Movie Rating System before supplying the elements requested below for a use case diagram of this system :	10
a) two actors	
b) 6 use cases	
c) two examples of includes use cases	

QUESTION 5 (10 MARKS)

What makes MobileApp development different and challenging when compared to WebApp development?	10
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QUESTION 6 (10 MARKS)

The MVC architecture has been selected as the most appropriate WebApp architecture for the Online Sentiment Based Movie Rating System . Explain	4
a) why this would be a good choice and	
b) how this architecture would work.	6

***** END OF PAPER *****